Site Standard CMS-001: Condition Monitoring Alarm Limits

1. Purpose and scope

This standard defines the alarm limits used by the online condition monitoring system for rotating equipment on site, and the response expected from operators and maintenance staff when a limit is reached. It applies to the P-Series pumps, M-Series motors and C-Series compressors.

Limits are based on the equipment manuals and on ISO 10816-3 vibration severity zones for medium-sized machines. Where a manual and this standard differ, the stricter value applies.

2. Definitions

- **Warning**: the measurement is outside the normal range. The equipment may stay in service with increased monitoring.
- **Alarm**: the measurement indicates likely damage. Inspection is required within 24 hours.
- **Trip**: an automatic protective stop, or a level at which the equipment must be stopped by the operator.

3. Measurement points

- Vibration: velocity in mm/s RMS, 10 to 1000 Hz, on each bearing housing. The monitoring system reports the highest of the measured directions.
- Temperature: bearing temperature for pumps, stator winding temperature (PT100) for motors, airend discharge temperature for compressors.
- Pressure: discharge pressure for pumps, delivery pressure for compressors.
- Current: motor current, compared with the rated current on the nameplate.

4. Alarm limits

Machine type	Measurement	Warning	Alarm
Pump (P-Series)	Vibration	4.5 mm/s	7.1 mm/s
Pump (P-Series)	Bearing temperature	80 °C	90 °C
Pump (P-Series)	Discharge pressure (low)	below 4.5 bar	below 3.5 bar
Pump (P-Series)	Motor current	105% of rated	115% of rated
Motor (M-Series)	Vibration	2.8 mm/s	4.5 mm/s
Motor (M-Series)	Winding temperature	130 °C	145 °C
Motor (M-Series)	Current	105% of rated	115% of rated
Compressor (C-Series)	Vibration	4.5 mm/s	7.1 mm/s
Compressor (C-Series)	Airend discharge temperature	105 °C	110 °C
Compressor (C-Series)	Delivery pressure (high)	8.5 bar	9.0 bar
Compressor (C-Series)	Motor current	105% of rated	115% of rated

Rated currents on site: P-1 27 A, P-3 54 A, M-1 56 A, M-2 82 A, C-1 67 A.

4.1 Evaluation rules

- A limit is considered reached when the median of the last three readings is beyond it. Single spikes are logged but do not raise a warning.
- Readings are taken every 5 seconds by the online system and stored for 30 days.
- A rising trend of more than 25% within 24 hours must be reported even if no limit is reached.

5. Required response

Level	Response	Ticket priority
Warning	Increase monitoring to daily manual readings. Plan an inspection within 7 days.	medium
Alarm	Inform the shift supervisor. Inspect within 24 hours. Stop the machine if the trend keeps rising.	high
Trip or safety risk	Stop the machine immediately and keep it isolated until the cause is found.	critical

5.1 Maintenance tickets

Every warning or alarm must be followed by a maintenance ticket in the maintenance system. The ticket must contain the machine ID, the measured values, the limit that was reached and the actions taken. Use priority **low** for observations without a limit violation.

5.2 Restarting after a trip

A machine that tripped on a protective limit may only be restarted after the cause has been found and corrected, and with the approval of the shift supervisor. Resetting and restarting without investigation is not allowed.

6. Review

This standard is reviewed every two years or after any failure that was not detected by the monitoring system.